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WORLD AS IT IS

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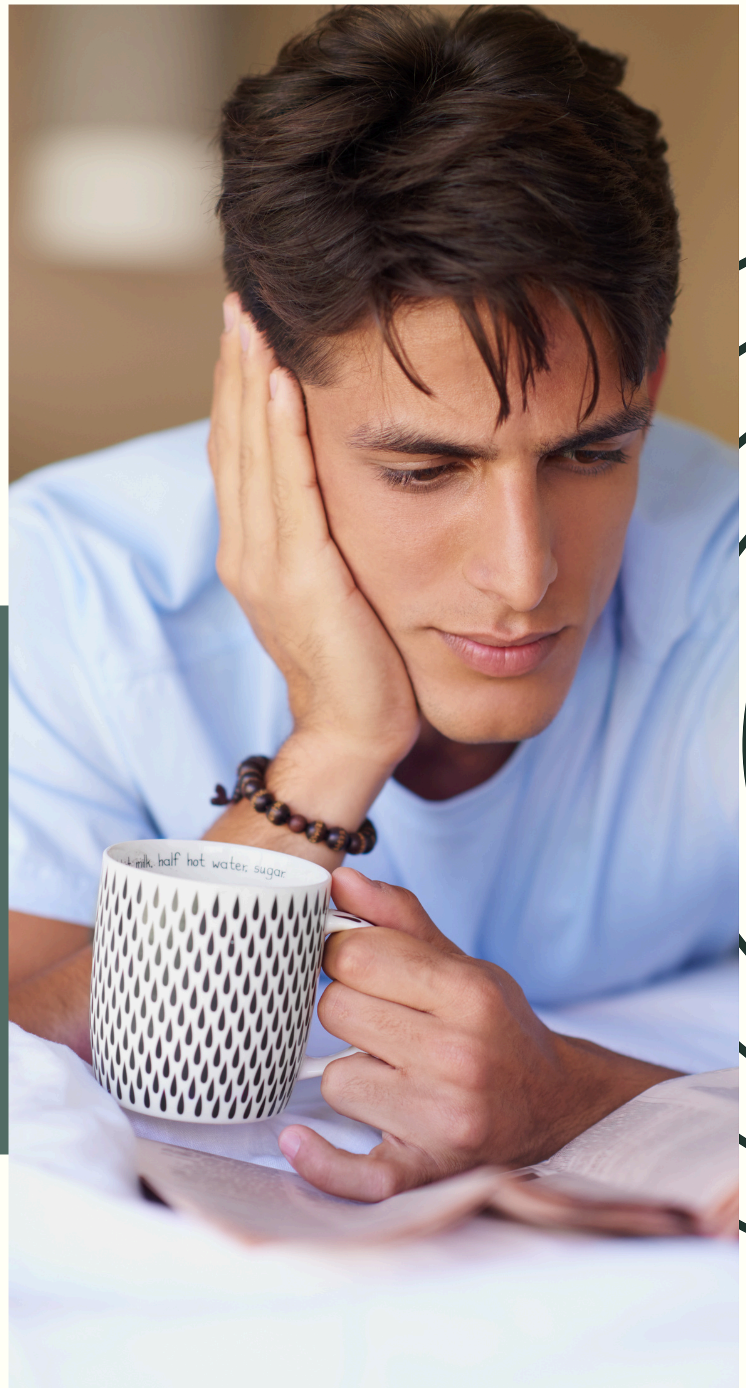
WHAT HAPPENED?

PHILOSOPHICAL AND MATHEMATICAL FOUNDATIONS

We had a philosophical foundation built over centuries, probability calculus, and mathematical formalism. Everything was under control.

CATASTROPHE WAS IMPOSSIBLE

Catastrophe was quantifiable and distant. So distant that it was impossible.
We were protected against catastrophes on the scale of once in 1000 years.
Everything was quantifiable, under control, in advanced and correct models.



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6 ASSUMPTIONS THAT FAILED

- **RISK IS MEASURABLE — UNCERTAINTY CAN BE CONVERTED INTO RISK**

This is the mother assumption. All other assumptions are its children. If this assumption is false — the entire project of mathematical finance is philosophically unjustified. CDOs, VaR, the Li copula — everything stands on this one foundation. Knight showed that this foundation does not exist.

- **THE NORMAL DISTRIBUTION DESCRIBES PRICE CHANGES AND LOSSES**

This is the technical assumption that killed. The normal distribution has thin tails. Reality has fat tails. This difference — which looks like a technical detail — means the model says that an event that destroys the system happens once in ten thousand years. In reality it happens once in twenty years. All of VaR, all regulatory capital, all AAA ratings stood on this error.

- **STABILITY IS THE NATURAL STATE — THE SYSTEM TENDS TOWARD EQUILIBRIUM**

The crisis arises endogenously. Not through an external shock. Through the very mechanism of a calm period that produces fragility. This is the assumption that makes models blind to risk as it grows — and screams when it explodes.

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6 ASSUMPTIONS THAT FAILED

- **CORRELATIONS ARE STABLE AND MEASURABLE FROM HISTORICAL DATA**

The direct technical cause of the 2008 crisis. The Li copula assumed low stable correlations between defaults. In the crisis correlations exploded to one. The entire CDO market — trillions of dollars — stood on this one parameter. When the parameter proved false — the market ceased to exist.

- **RISK IS EXOGENOUS — THE MODEL MEASURES IT BUT DOES NOT CREATE IT**

This is the assumption that makes regulations amplify crises instead of dampening them. When all banks use the same model — the model synchronizes their behavior. Synchronized behavior creates systemic risk that the model cannot see because it assumes risk is external. This is the tragic paradox — the better the model the more dangerous it becomes when widely applied.

- **WE KNOW THE SCALE AND PROBABILITY OF CATASTROPHE: ONCE IN 1000 YEARS**

This is perhaps the most absurd concept. How could one have succumbed to such an illusion?

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THE WORLD OF CRISIS IS NOT NORMAL

GAUSSIAN COPULA LI CAN NOT WORK

FAT TAILS — PHASE TRANSITION

In physics a phase transition is the moment when a system suddenly changes its global state — water freezes, steel demagnetizes. It does not happen gradually — it happens abruptly when the temperature crosses a threshold.

In finance this threshold is collective belief. As long as everyone believes, the system is in the normal phase. Small correlations, normal distribution, catastrophe impossible. When belief disappears — an instantaneous phase transition to the crisis phase. Correlations jump to 1, everyone sells simultaneously, asset values plummet vertically.

This is not a gradual deterioration. It is a discontinuity. And precisely because continuous models — based on the normal distribution — cannot describe this. The normal distribution has no discontinuities. It has smooth tails that taper gently.

MANDELBROT (1963)

"The Variation of Certain Speculative Prices"

He saw two things that the normal model did not predict. First — the middle was too high. On days with very small changes, more than the normal distribution predicts. Second — the tails were too fat. On days with extremely large changes, many times more than the normal distribution predicts. This is precisely leptokurtosis — too high a peak and too fat tails simultaneously.

No one listened to him.

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WHAT IS A CRISIS?

NATURE

This is a more objective concept — rare occurrences of a state of nature. But it is also subjective — it concerns us when it affects us. Can it be modeled statistically: once in 100 years? This is currently being challenged: "Stationarity is dead." The entire concept of statistical modeling is being called into question.

ECONOMY

It is very difficult to define what a crisis is in the world of economics.

Most likely one would have to say that it is a collapse of the world of value.

But what is value?

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HISTORICAL CRISES

- ## TULIP MANIA 1637

In 1636-1637 Dutch tulip bulbs reached prices equivalent to an Amsterdam townhouse, driven by speculative futures contracts and fear of missing out. When the market collapsed in February 1637 within days, prices fell 99% and the contracts became worthless — a purely psychological bubble that burst the moment new buyers willing to pay ever higher prices ran out.

- ## SOUTH SEA BUBBLE 1720

The South Sea Company had almost no real trading activity, yet its shares rose from 100 to over 1000 pounds in a single year, driven by government backing and the narrative of future profits. When insiders began quietly selling in the summer of 1720 the price collapsed back to 100 pounds, ruining thousands of investors including Isaac Newton, who reportedly said he could calculate the motion of heavenly bodies but not the madness of men.

- ## BANKING PANICS 19TH CENTURY

Throughout the 19th century, financial crises struck roughly every decade through the same mechanism — fractional reserve banking meant that a single rumor about a bank's health could trigger a run, causing even solvent banks to collapse as they could not pay all depositors at once, with each failure spreading panic to the next. The panics of 1837, 1873, 1893, and 1907 shared one common denominator: no lender of last resort, no deposit insurance, and a system held together by trust alone — making its collapse a self-fulfilling prophecy, ultimately leading to the creation of the Federal Reserve in 1913.

- ## WALL STREET CRASH 1929

Stocks bought on 90% borrowed capital collapsed in October 1929, triggering margin calls that forced mass selling, bankrupting over 9000 banks and shrinking the money supply by one third — while the Federal Reserve raised interest rates to defend the gold standard, turning a crash into the Great Depression with 25% unemployment and a 30% fall in GDP. The indirect consequence was Hitler's rise to power in Germany, where the crisis struck with particular force.

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THREE DIFFERENT PROBLEMS EMERGE

- **WHAT IS VALUE?**

Does objective value exist?
Who or what determines value?
Can we approximate natural value? S
hould the valuation of value be controlled?
What are the components of value?

- **SPECULATIVE TOOLS**

People create speculative tools through which they see a different world of value. These tools are often sophisticated, and therefore attractive. People enjoy deceiving themselves when these tools point to a higher value than the real one.

- **PANIC AT THE MOMENT OF TRUTH**

Market participants react with panic when the world they believed in collapses. Such a reaction causes a deeper crisis than there would have been had they reacted normally.

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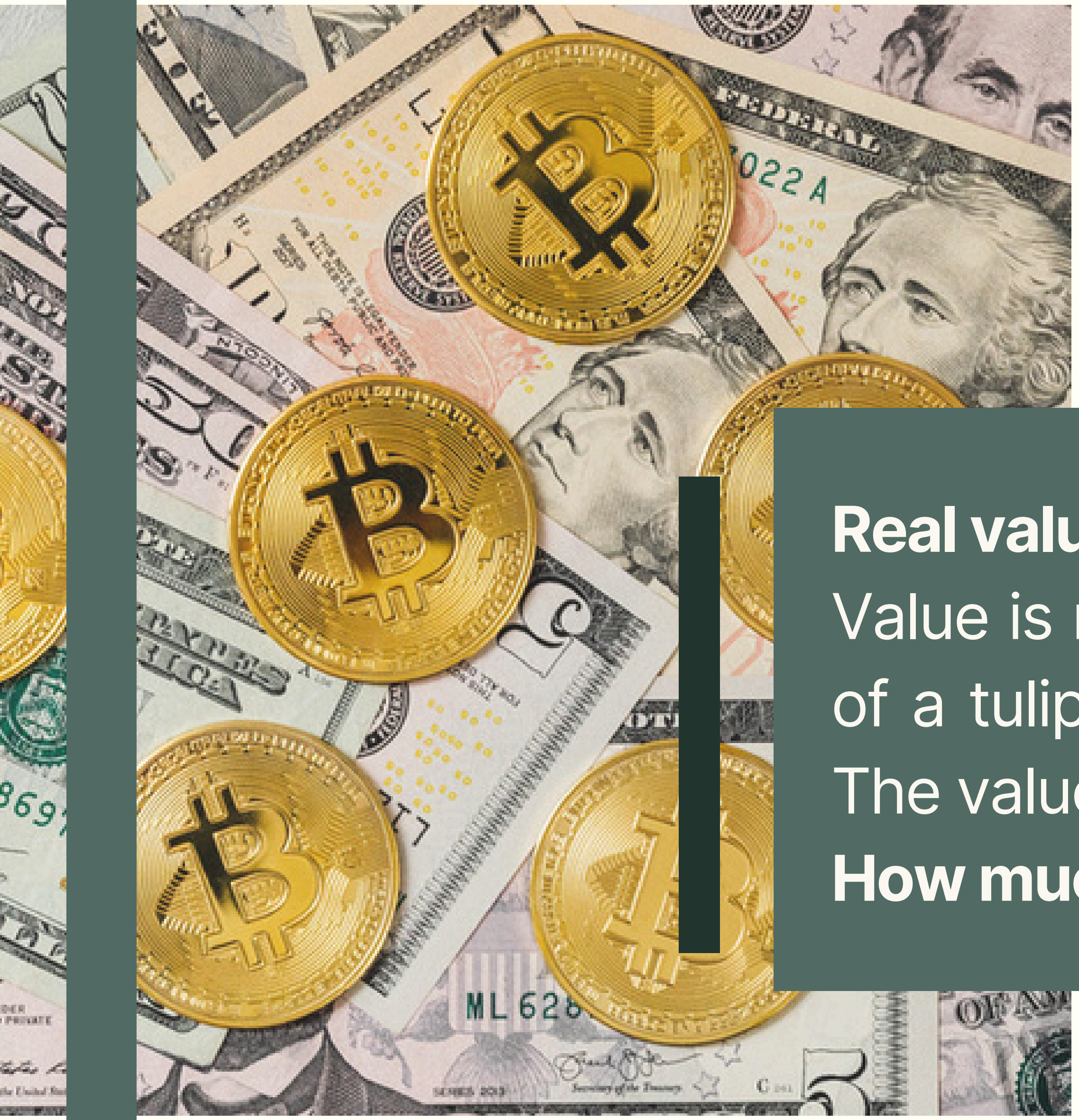
DOES OBJECTIVE VALUE EXIST?



Real value does not exist as an objective fact.
Value is not a property of a thing like mass or temperature. The mass of a tulip is measurable independently of what people think about it. The value of a tulip does not exist without the person who assigns it.
How much is a Van Gogh painting worth?

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RISK AS THE INVISIBLE COMPONENT OF VALUE

REDUCED PORTFOLIO VALUE

If we have a portfolio value of for example 100 million, we must assume that some loan customers will not repay their credit. This requires a revaluation of the portfolio by the portion that will not be repaid. This portion is credit risk.

THE MOMENT OF THE TEST

In normal periods of time it is relatively straightforward to calculate the level of credit risk through the ECL process. The market behaves in a stable manner which facilitates this estimation. However it is never known how much we will lose at the moment of a crisis, although models and the regulator claimed that we were able to determine this at a probability level of once in 1000 years.

VALUATION OF VALUE - ABSTRACT

- **RISK IS CRUCIAL PART OF VALUE**

Risk is invisible. It is not known how to value it. statistical models come to the rescue with their assumptions about randomness. They are difficult to challenge, because no one knows what the risk is.

- **MODEL CALCULATE ANYTHING**

Models can calculate whatever we want. Statistics is flexible. It is possible to arrive at results that we desire. And we always desire results that yield greater profits.

- **LET US COMPLICATE THIS FURTHER**

Securitization allows the object of risk to be rotated along a different axis, in such a way that a large part of that risk disappears. As if it did not exist. It is an obvious speculation that the further an instrument is from the object of risk the less visible that risk becomes. It is therefore easy not to see it.

- **LET THE GAME CONTINUE**

Since we do not know what value is, because it is intersubjective, let us go further. Let us bet on what value is and place trillions of dollars on those bets. Why not?
It is only when real people's wealth is at stake that they become afraid.

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JUDGES OF VALUATION

RATING AGENCIES

Moody's, Standard & Poor's, Fitch — three agencies that controlled virtually the entire ratings market. A duopoly of Moody's and S&P — each holding approximately 40% of the market. Fitch had the remaining 20%. And they were regulatorily privileged. The SEC — the American regulator — designated them as Nationally Recognized Statistical Rating Organizations — NRSROs. Only their ratings were recognized for regulatory purposes.

MONOPOLISTIC POWER

Pension funds, banks, insurers — all had regulatory requirements to hold only instruments with a specified rating. AAA from Moody's or S&P — and you can buy. No rating or rating below BBB — you cannot. This gave the agencies monopolistic power. Without their AAA an instrument could not be sold to the largest investors.

RATING SHOPING

The mechanism that was at the heart of the problem. An investment bank was building a CDO. Before formally submitting a rating application — it held informal conversations with the agencies. It asked — if I build a CDO with this structure, with these assets, with this division into tranches — what rating will the senior tranche receive? The agency responded — with these parameters it will receive AAA. The bank said — and if I change the structure in this way — will it still receive AAA? The agency responded — yes, with this change still AAA. This was an iterative optimization process. The bank optimized the CDO structure to obtain the maximum rating at minimum cost — minimum overcollateralization, minimum equity tranche. The agency was in effect helping to design the instrument to receive AAA. And then issued the AAA rating. This is the exact opposite of independent risk assessment.

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LOSS OF A FUNDAMENTAL PROPERTY OF THE SYSTEM

At the moment of crisis not only is there collective panic, but another and more fundamental property of the system is lost — its ability to balance itself through the independent perspectives of different participants. Now everyone is in panic, and so this mechanism no longer exists. The system no longer shows market valuations, it has stopped delivering market information.

THIS IS VERY IMPORTANT — THE SYSTEM LOSES ITS LEGITIMACY, ITS FOUNDING MYTH, THAT IS EQUILIBRIUM AND VALUATION AS THE EXCHANGE OF INFORMATION IN THE MARKET.

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CRISIS – DEEP VIEW

ITS DANGEROUS

This is why crises are so destructive even when "nothing real has changed" — factories stand, people exist, technologies are there. Only the system of collective beliefs that organizes cooperation between people has changed. And without this system factories do not produce, people do not cooperate, technologies are not implemented.

Money, credit, valuation — these are social technologies that exist only in the collective imagination. A crisis is a failure of this technology. **And just like a failure of physical infrastructure — a bridge, an energy grid — a failure of social infrastructure paralyzes everything that depends on it.**

SO WHAT IS A FINANCIAL CRISIS ONTOLOGICALLY?

It is the moment in which the system of collective beliefs about value loses its internal coherence and cannot coordinate around a new equilibrium point without external intervention.

This is not a return to reality after a period of illusion — because there is no objective reality in the financial world in that sense. It is a transition from one intersubjective reality to another, through a zone of chaos in which the old reality no longer works and the new one has not yet formed.

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THREE TYPES OF REALITY ...

WHAT ARE THREE TYPES?

Objective — exists independently of what we think, like gravity.

Subjective — exists only in one mind, like pain.

Intersubjective — exists only when many people simultaneously believe in it, like money, law, nation, a joint-stock company.

MOST FINANCIAL VALUES ARE INTERSUBJECTIVE REALITY

A share is worth one thousand zloty not because it has that value inherently, but because enough market participants simultaneously agree that it is worth that much and act in accordance with that belief. **A crisis is the moment when this intersubjective consensus collapses.** Not gradually — violently, because beliefs are networked with each other. When enough participants change their minds, the change becomes self-fulfilling. Everyone sees that others are selling and sells themselves — not because they changed their fundamental assessment but because their assessment of what others think has changed. This is precisely Keynes's beauty contest.

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INTERSUBJECTIVE REALITY

AT THE MOMENT OF CRISIS

NO TRUST

The crisis fully reveals the intersubjectivity of value. No one believes in the value of assets. Banks do not want to finance each other because no one knows what the value of assets is. This happens because there is no objective value.

NEW INTERSUBJECTIVITY

A player is needed who will create a new kind of certainty. They must possess: unlimited resources, credibility, absence of short-term profit motive. Only then will everyone believe them, believe in the new intersubjective state and calm will return. Only **the state or a central bank** can provide this.

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INTERVENTION OF A LARGE PLAYER

“WHATEVER IT TAKES” MARIO DRAGHI

Draghi did not buy any bonds at the moment he said it. He only changed the beliefs of market participants about what he would do. This changed their behavior. This changed prices. This confirmed the new equilibrium. A new intersubjective reality arose from words — because the words were spoken by someone with sufficient credibility for everyone to believe them.

This is sovereign power in a non-philosophical sense. The state can create financial reality through a declaration of will — but only when it is sufficiently credible.

MORAL HAZARD

If the state always intervenes in a crisis — market participants begin to count on it. They take greater risks because they know that in the event of a catastrophe they will be rescued. Economists call this moral hazard.

Bernanke, Greenspan, Draghi — they all knew about this dilemma. There is no good way out. You do not intervene — the crisis deepens and destroys the real economy. You intervene — you rescue the system but you are building the next bubble.

UNCERTAINTY CANNOT BE CALCULATED

FRANK HYNEMAN KNIGHT

"Risk, Uncertainty and Profit", 1921.

Knight introduced a distinction that is fundamental and which mainstream economics consistently blurs: uncertainty vs. risk.

RISK

A situation where you do not know the outcome — but you know the probability distribution of possible outcomes. A dice roll. You do not know whether six will come up — but you know the probability is one sixth. You have complete information about the structure of randomness.

UNCERTAINTY

A situation where you do not know the outcome — and you do not know the probability distribution. You have no basis for assigning numbers to possible outcomes. Will a new product achieve market success? Will this technology transform the industry? Will there be a war in ten years?

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WHAT IS THE TRUE CAUSE OF CRISES?

NOT THE VIRTUALITY OF VALUE ITSELF — FOR THAT IS UNAVOIDABLE

The cause is that people forget about this virtuality. During periods of growth they begin to treat market values as objective and permanent. They build on them layer by layer — borrowing against them, issuing derivative instruments, pricing future profits assuming the trend lasts forever. A pyramid of mutual expectations is created, ever more detached from any real foundation.

A crisis is not the moment when value disappears — because it was never fully real in the first place. A crisis is the moment when everyone simultaneously remembers this

HYMAN MINSKY

"The Financial Instability Hypothesis"

Hyman Minsky described this as an inseparable feature of financial capitalism — stability breeds instability. A long period of calm causes market participants to take ever greater risks, because risk appears small. Until the moment it does not.

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MINSKY PHASES

RISK IS INSIDE THE SYSTEM - SYSTEMIC RISK ACUMULATING

- **STEP ONE — AFTER THE CRISIS.**

After every financial crisis market participants are cautious. They remember the losses. Banks have conservative lending standards. Companies have low leverage. Investors hold large cash buffers.

- **STEP TWO — THE CALM PHASE**

Growth continues. Companies earn. Loans are repaid. Banks record no losses. History says — it is safe. And here the mechanism begins. Minsky said that expectations are shaped by experience. If there has been no crisis for five years — participants begin to think that a crisis is unlikely. If for ten years — they begin to think the business model is solid. If for fifteen years — they begin to think that previous crises were exceptions.

- **STEP THREE - PONZI PHASE**

Hedge financing — safe. Cash flows from operations cover both interest and principal repayment. Even when revenues fall — the company survives.
Speculative financing — risky. Cash flows cover interest but not principal repayment. The company must roll over debt — take new loans to repay old ones. Works when credit markets are open.
Ponzi financing — very risky. Cash flows do not even cover interest. The company survives only if the prices of assets it holds rise — to sell assets and repay debt. Depends entirely on price growth.

- **MINSKY MOMENT**

The Minsky Moment is the inflection point — the moment when the system transitions from the accumulation of risk to its sudden realization.

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CRITIQUE OF THE FINANCIAL SYSTEM ...

HAYEK — NOBEL LECTURE 1974, NOBLE PRIZE IS A PROBLEM ITSELF

- ### FIRST THESIS — ECONOMICS IS NOT PHYSICS

Hayek draws a fundamental distinction between natural sciences and social sciences.
In physics — the phenomena you study are measurable. You can control the conditions of the experiment. You can collect data that is representative of the mechanism you are interested in.
In economics — the phenomena you are interested in are too complex to measure in a way that corresponds to their real structure.
And here Hayek introduces the key concept — phenomena of organized complexity.

- ### SECOND THESIS — THE ERROR OF SCIENTISM

Hayek introduces the concept of scientism. Scientism is not science.
It is the mistaken imitation of the method of natural sciences in fields where that method is not appropriate. Economists — says Hayek — fell in love with the methods of physics. With equations, models, numbers. Because physics is prestigious. Because numbers look precise. Because mathematical models can be published in prestigious journals.
But mathematical precision does not correspond to precision of knowledge about reality.

- ### THIRD THESIS — WHAT ECONOMISTS SHOULD DO

Economists should practice negative knowledge. Instead of saying what should be done — saying what should not be done.
Instead of building models that predict — building understanding that warns.
Instead of asking — what policy will produce 2% GDP growth? Asking — what policies lead to catastrophes and why?

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CRITIQUE OF THE FINANCIAL SYSTEM

BLAUG AND ROMER

- ### BLAUG 1999 - CRITIQUE OF "FORMALIST REVOLUTION"

Blaug said that economics after the 1950s underwent a formalist revolution. Arrow-Debreu, game theory, econometrics — mathematics became an end in itself rather than a tool.
His key phrase — economists practice "rigor mortis" instead of "rigorous economics." Literally — the rigor of death instead of rigorous economics. The formal precision of a dead science instead of a living understanding of reality.

- ### ROMER 2015 - "MATHINESS"

Paul Romer — Nobel laureate 2018 — wrote a provocative article "Mathiness in the Theory of Economic Growth", American Economic Review, 2015. He used precisely the word "mathiness." He described how mathematical symbols are used in economics to hide a lack of content or to disguise ideological assumptions. Mathematics as rhetoric, not as science.

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NASSIM TALEB

BLACK SWANS

- **EPISTEMIC THESIS**

We do not know as much as we think we know. Our knowledge of the future is fundamentally limited — not by a lack of information that can be supplemented, but structurally. Taleb distinguished two types of uncertainty. Risk — you know what the possible events are and what probabilities they have. You can calculate expected values, insure yourself, optimize. Knightian uncertainty — you do not know what events are possible. You have no distribution. You cannot calculate. And he said that most important events in history — and in finance — belong to the second category. But people and models treat them as the first. This is a direct reference to Knight 1921.

- **BLACK SWANS**

Three characteristics:
First — it is beyond the normal spectrum of expectations. Nothing in the past indicated it was possible.
Second — it has an extreme impact.
Third — after the fact it is rationalized as predictable. We construct a story that "this was foreseeable".

- **NARRATIVE ILLUSION**

People understand the world through narratives. And this is the problem. Narrative simplifies, linearizes, imposes causality and meaning on events that were chaotic and random. History is written by those who survived — survivorship bias. We do not see all the trajectories that could have happened, we see only the one that did happen. And we construct a story of why that one was inevitable.

LESSONS?

LESSONS – REGULATOR RESPONSE FOR CRISIS 2008

- **EXPECTED CREDIT LOSS - IFRS 9**

IFRS 9 — International Financial Reporting Standard 9 — replaced IAS 39 in 2018 and fundamentally changed how banks account for credit losses, shifting from an incurred loss model to an expected credit loss model. Under the new standard banks must recognize losses not when they actually occur, but when they are expected — based on forward-looking information, macroeconomic scenarios, and probability-weighted outcomes across the lifetime of a financial instrument. This was a direct regulatory response to the 2008 crisis, where the old standard was criticized for recognizing losses "too little, too late," amplifying the crisis rather than cushioning it.

- **IRB CORRECTED**

After the 2008 crisis, regulators recognized that banks using Internal Ratings-Based models had systematically underestimated risk by calibrating their models on calm historical periods, producing risk-weighted assets that were too low and capital buffers that were too thin. Basel III and subsequent reforms introduced output floors — requiring that IRB-based capital requirements cannot fall below a certain percentage of the standardized approach, directly limiting the ability of banks to use models to minimize regulatory capital. Additionally, supervisors introduced more rigorous model validation requirements, stress testing, and the Targeted Review of Internal Models — TRIM — to ensure that internal models were not used as tools for capital optimization rather than genuine risk measurement.

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WHY 1000-YEARS CRISIS HAPPENS?

WERE WE SAFE AT THE LEVEL: 1/1000 YEARS

- **RANDOMNESS IS NOT STATE OF NATURE**

It is merely a concept, an approach — a way in which we look at the world. Real events escape this concept.

- **THE WORLD IS NOT NORMAL**

The world is not normal, therefore one cannot read the probability of catastrophe: once in 1000 years from the normal distribution (quantile 0.999)

- **SYSTEM IS NOT STATIONARY**

And it never was. The economic system is by assumption non-stationary — it is the result of human developmental activity.
It is therefore impossible to find a distribution from which one can read the 0.999 quantile (once in 1000 years)

- **SYSTEMIC FACTOR**

The behavior of market participants at the moment of crisis depends primarily on the systemic factor.
One cannot read probabilities from a stable normal distribution.

- **RISK IS WITHIN**

Risk is within the system, not outside it as in hydrology (although increasingly less so even there). The main risk is precisely the behavior of market makers: banks and regulators, who are inside the system. We therefore cannot look at risk from above and model it, because we do not account for the risk that we ourselves create.

- **MODELS CREATE RISK**

Models create risk. They appear to describe risk from the outside, but in reality they are part of the risk. Their universality and standardization only amplified the risk.

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ILLUSIONS ARE ATTRACTIVE

All illusions are attractive, especially when in normal conditions they look like truth, and when they also bring benefits, the benefits prevail.

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INSTRUMENTALISM WITHOUT ACCOUNTABILITY

Unreflective instrumentalism, without searching for true meanings, is only possible when there is no accountability for decisions.

A person in their own decisions cannot afford this.

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INTERNAL CONTRADICTION

On one hand we enthusiastically develop the world and this is our main goal. Development, movement is the realization of our freedom.

On the other hand we expect the world not to change, so that we can model the risks.

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RISK AS A PARAMETER

Market makers on one hand model risk, looking at it from above as something external.
On the other hand they treat risk models and risk-limiting regulations as a parameter of their game. They optimize their actions around these parameters. They create risk. All calculations of once in 1000 years are merely a parameter, and then they create risk by generating speculative solutions that help maximize profits.

BRIAND - KELLOG AGREEMENT (1928) - FROM THAT MOMENT, THERE IS NO WAR

BUT STATES TREAT THAT REGULATION AS PARAMETER

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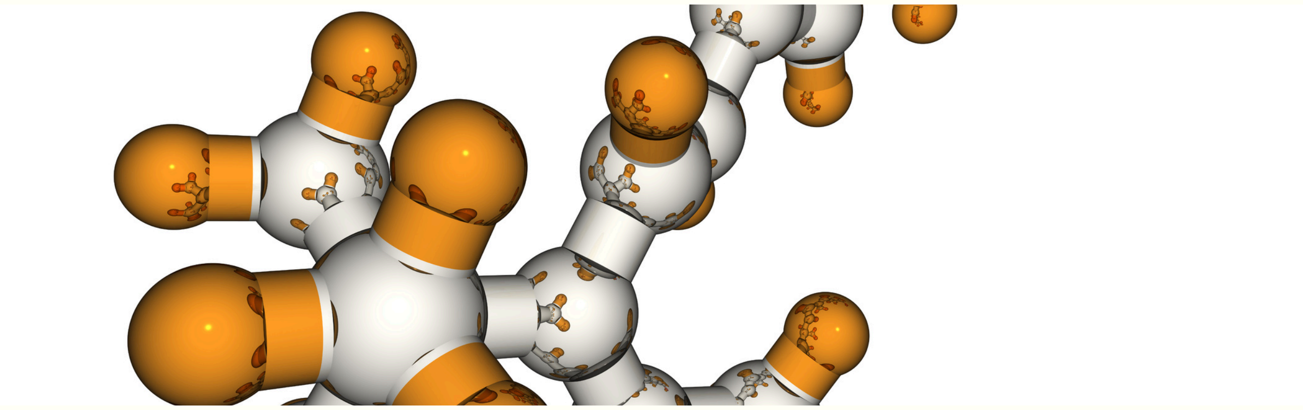
FUTURE?

BREIMAN (2001) - TWO CULTURES OF MODELING

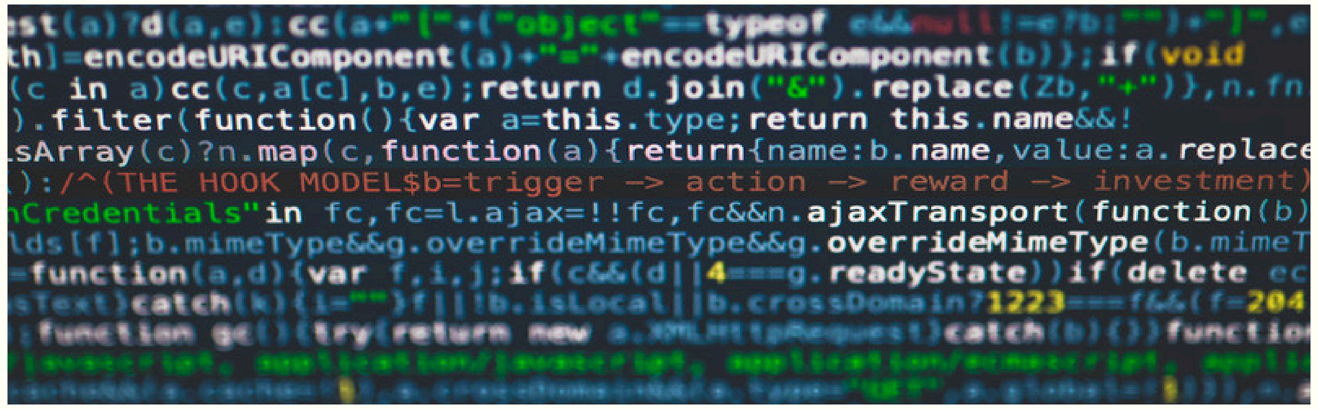
CULTURE ONE — DATA MODELING

You assume there is a true mechanism generating the data. You build a model of this mechanism. You ask — is my model true? Do the parameters make sense? Does the structure of the model correspond to the structure of reality?



CULTURE TWO — ALGORITHMIC

You are not interested in the mechanism. You are interested in prediction. A black box is acceptable if it works. You measure only — is the answer correct. **Modern neural networks are culture two in its most radical form. More radical than Breiman could have imagined in 2001.**



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THANK YOU

FOR YOUR ATTENTION

PRESENTED BY
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17 MARCH 2026